Day - 1 (Web Fundamentals)

24 October 2024 16:26

- **Every Website Have Two Parts:** Frontend and Backend are key terms in web development. The frontend is what users see and interact with, while the backend is how everything works behind the scenes. Both sides need to work together seamlessly to ensure a functional website.

  - **Frontend:** The frontend is the part of the website users directly interact with. This includes the design, menus, text, images, videos, and overall layout. The main languages used for frontend development are HTML, CSS, and JavaScript. Responsiveness and performance are the two main objectives of the Front End. The developer must ensure that the site is responsive i.e. it appears correctly on devices of all sizes no part of the website should behave abnormally irrespective of the size of the screen.

  - **Backend:** The backend is the server side of the website. It manages data and ensures everything on the frontend works properly. Users don't see or interact directly with the backend; it's the behind-the-scenes functionality. The parts and characteristics developed by backend designers are indirectly accessed by users through a front-end application. Activities, like writing APIs, creating libraries, and working with system components without user interfaces or even systems of scientific programming, are also included in the backend.

- **HTML:** It is a markup language and it is use to structure the web page. Also responsible for content.

- **CSS:** It is cascading style sheet which is used for styling the webpage.

- **JavaScript:** It is programming language which add functionality and behaviour in the webpages.

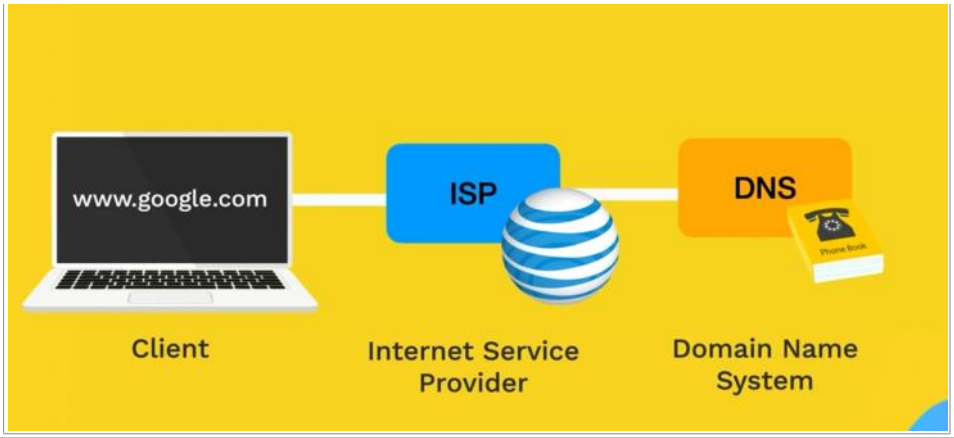
  - **Real Life Example:**
    A building have structure and we can colour it to look beautiful and add functionality like escalator, etc.

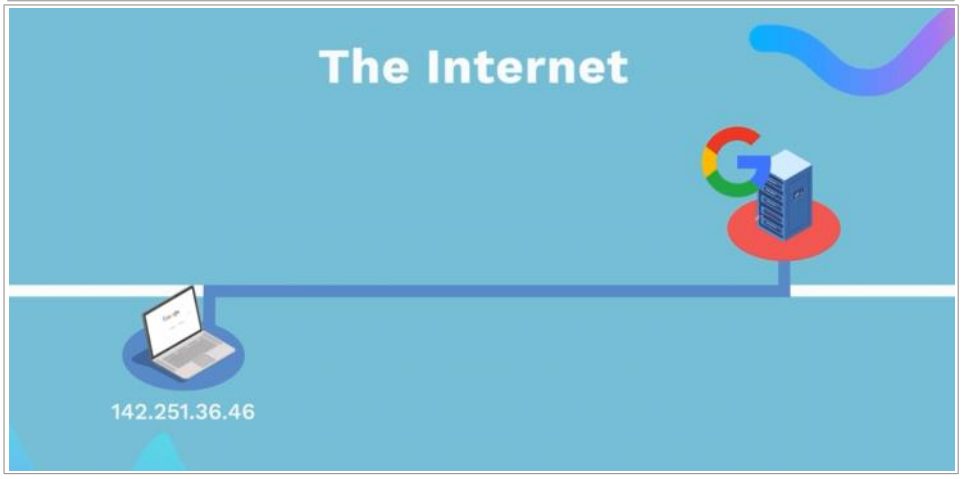
- **How Internet Works:**

  - **The Internet:** It is long piece of wire which is spread all under the ocean and it connected one computer to another.

  - When client search the website then the browser send the message to ISP (Internet Service Provider)
  - ISP rely this message to DNS (Domain Name System), it is phonebook which search the IP of the websites.
  - After searching it send message back to the client.

Search IP Addresses: nslookup.io
Subraine Cable: Submarine Cable Map

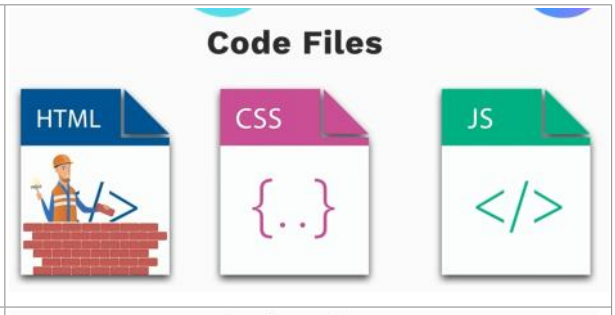




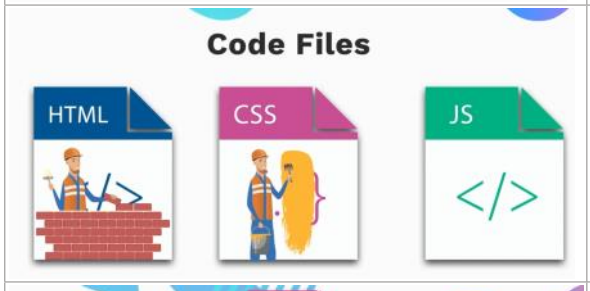
★ How Website Render On Browser:

1

2

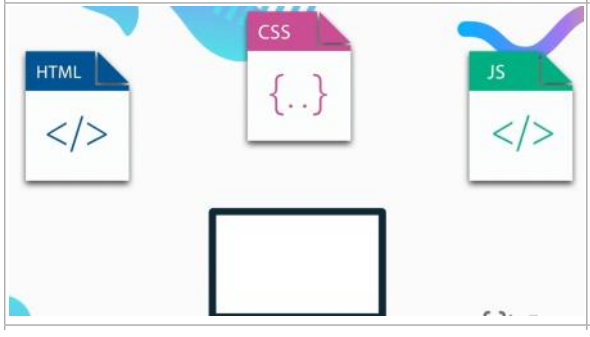


3

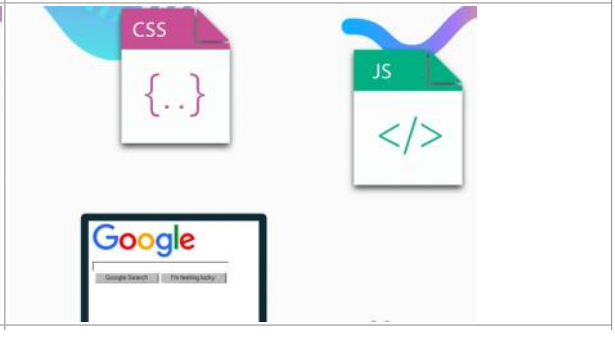


4

5



6



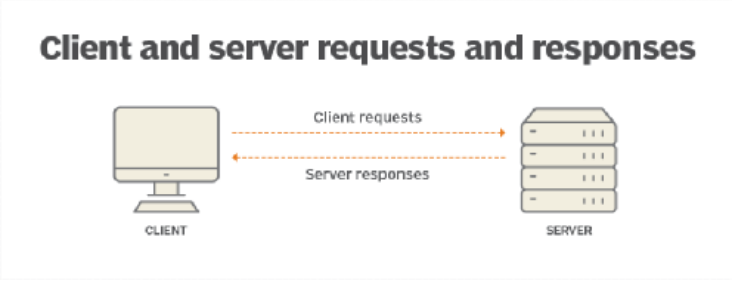
7

8

★ How Website Works:

- Let's Type This URL In the browser:
- http://www.abc.com : This is URL (Uniform Resource Locator) and it is way to locate resources on the internet. Resources can images, audio, video, etc...

Now Browser involves two pieces here:



- The client send request to server and server send the resources or response to client.
- This message is formatted based on the HTTP And HTTPs Protocol. It is used to communicate between client and server.

- **HTTP:** It is a plain textual language for communicating.
- **HTTPs:** HTTP + encryption.

Example:

Request Message:(HTTP Request)

GET /index.html HTTP/1.1

Host: www.abc.com

Accept-Language: en-us

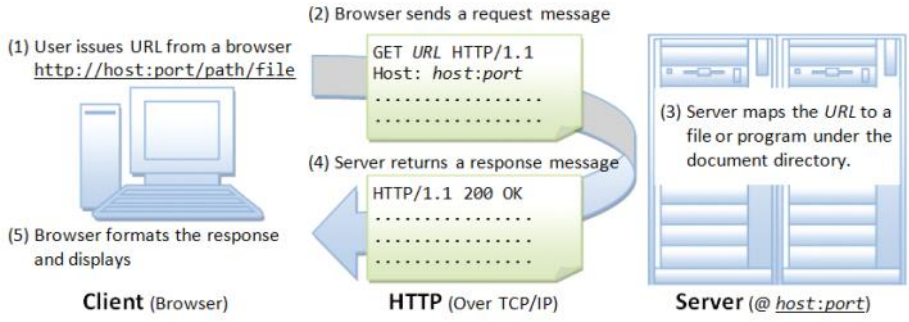
Response Message:(HTTP Response)

HTTP/1.1 200 OK

Date: 24 October 2024 16:26

Content-Type: text/html

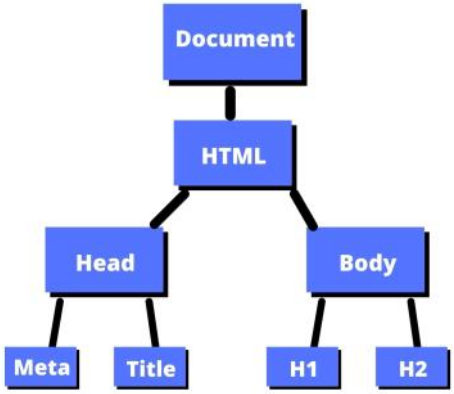
.....HTML Document.....



Status Code

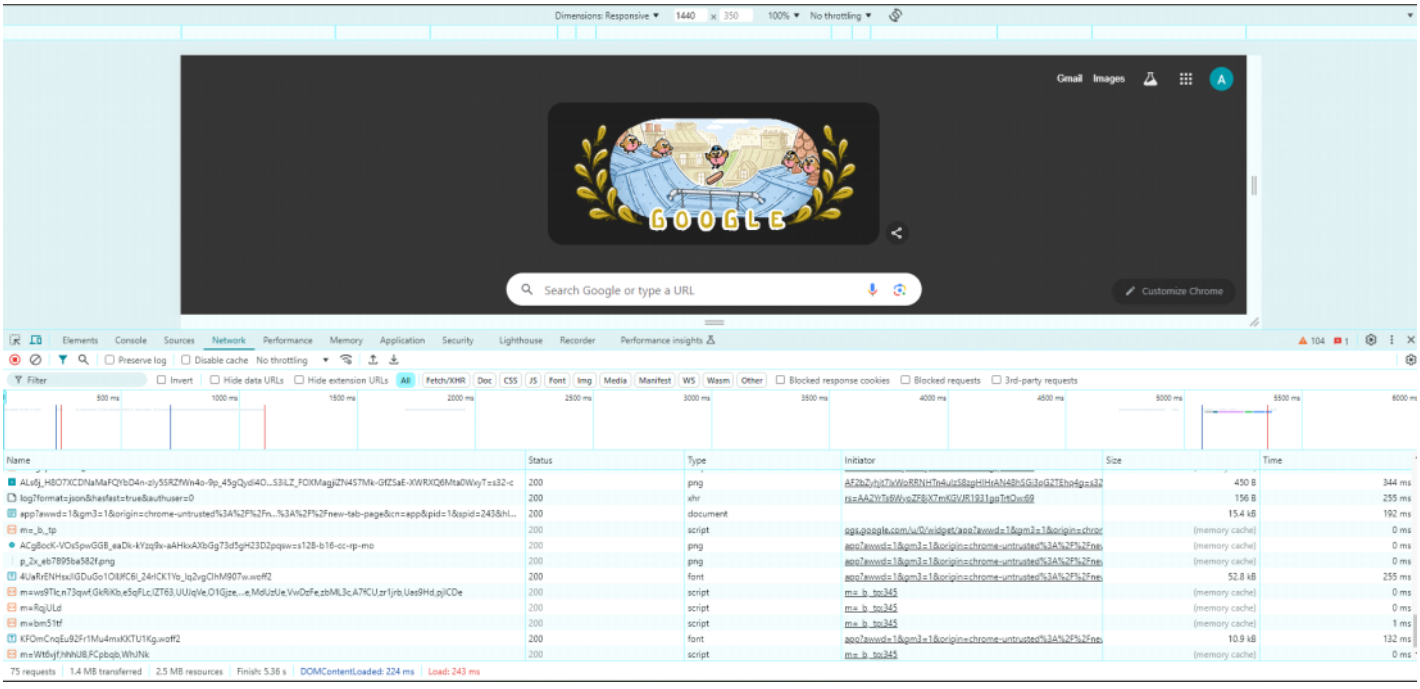
★ DOM:

- **Document Object Model:** It represents the objects or elements in our HTML document.
- **This elements are**: paragraphs, images, fonts and other stuffs.
- For each resources the browser sends separate HTTP request to the server to fetch that resource.
- Many requests send it parallel for fast execution.



★ Inspecting HTTP Request And Response:

- Open the inspect, by right click on the web page and select inspect. Shortcut Key **(Ctrl + Shift + i)**.
- Now go to network.
- Press the refresh button.
- Now these all are the HTTP request that chrome send to the google.
- There are total request left bottom corner with total size in KB or MB.
- You filter the requests and also get more details about that request by click on that request.



★ Validating Web Pages:

Copy Address Link And Paste It To Browser:

- For HTML: HTML Validator
- For CSS: CSS Validator